



EXPLOITS UNIVERSITY

BSc INFORMATION COMMUNICATION AND TECHNOLOGY

BICT 3202 · OBJECT-ORIENTED ANALYSIS AND DESIGN

WEEK 1

Assignment: Business Needs, Software Needs and Object Technology

PROGRAMME

BSc Information Communication and Technology

COURSE CODE / YEAR

BICT 3202 · Year 3

LECTURER

Francis Fweta — ICT Lecturer

DELIVERY

2h Lecture · 1h Tutorial · 1h Practical · 10h Independent Learning

Prepared by Francis Fweta · Exploits University · Week 1 of 16



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ASSIGNMENT INSTRUCTIONS

- Answer ALL questions in Sections A, B and C.
- Read the Week 1 module and your lecture notes before attempting each question.
- Use real-world examples from business, banking, or university systems.
- Cite ISO/IEC/IEEE 29148:2018 where relevant.
- Submit by the date given by your lecturer. Total: 50 marks.

Section A — Short-Answer Questions (20 marks)

Question A1

Define a business need. [2 marks]

Question A2

What is a business problem? [2 marks]

Question A3

Differentiate between a business problem and a business need. [3 marks]

Question A4

What is a business objective? [2 marks]

Question A5

Define a stakeholder. [2 marks]

Question A6

What is a functional requirement? [2 marks]

Question A7

What is a non-functional requirement? [2 marks]

Question A8

What is a business rule? [2 marks]

Question A9

Define object technology. [2 marks]

Question A10

What is an object? [1 marks]

Section B — Application Questions (20 marks)

Question B1 (10 marks)

A hospital has patients waiting several hours before being attended to.

Identify:

- (a) The business problem [2 marks]
- (b) Two business needs [2 marks]
- (c) Two business objectives [2 marks]
- (d) Five stakeholders [2 marks]
- (e) Five functional requirements [2 marks]

Question B2 (10 marks)

A university wants an online examination management system.

Identify:

- (a) The business need [2 marks]
- (b) Five stakeholders [2 marks]
- (c) Five functional requirements [2 marks]
- (d) Five non-functional requirements [2 marks]
- (e) Five candidate objects [2 marks]

Section C — Case Study (10 marks)

'A retail organization currently records sales using handwritten receipts. Management wants to introduce a computerized sales management system.'

As an object-oriented systems analyst, explain how you would move from understanding the business need to identifying potential objects for the proposed system.

Your answer should discuss:

- Current business problem;
- Business objectives;
- Business needs;
- Stakeholders;
- Requirements elicitation;
- Functional requirements;
- Non-functional requirements;
- Business rules;
- Candidate objects;
- State and behaviour of the candidate objects.



Marking Scheme

Section	Content	Marks
A	Short-answer questions (10 questions)	20
B	Application questions (10 + 10)	20
C	Case study	10
	TOTAL	50

Submission & Academic Integrity

- Submit a typed document (or neatly handwritten, as directed by your lecturer).
- This assignment is individual work. Plagiarism or copying will attract a penalty under university regulations.
- Cite any sources you consult using the format used in the Week 1 module references.